



**Q1.** State two characteristic properties of nuclear force.

- (i) We know that nuclei consists of tightly packed protons, thus to keep these protons together the nuclear force has to beat the Coulomb repulsion. At short distances nuclear forces is stronger than the Coulomb force, so nuclear forces are the strongest forces in nature.
- (ii) The nuclear force does not depend at all on the particle charge, e.g. it is the same for protons and neutrons. Nuclear forces are independent of charge.

**Q2.** How does the angle of minimum deviation of a glass prism vary, if the incident violet light is replaced with red light?

We know that deviation produced by prism is given by  $\delta = A(\mu - 1)$

$\mu$  is refractive index of the prism of mean ray( yellow colour ).

$\mu$  depends of wave length by this formula  $\mu = A + \frac{B}{\lambda^2}$

VIBGYOR – Wave length increases from Violet to Red

So Wavelength of Violet is less than Wave length of Red.

Hence Refractive index of Prism for higher for Violet than Red.

So  $\delta_{\text{violet}} > \delta_{\text{red}}$  therefore when incident violet is replaced with red light, the angle of minimum deviation of a glass decreases.

**Q3.** The instantaneous current and voltage of an a.c. circuit are given by

$$i = 10 \sin 300t \text{ A and } v = 200 \sin 300 t \text{ V.}$$

What is the power dissipation in the circuit?

Ans. We know that instantaneous current and voltage is given by  $i = i_0 \sin \omega t, V = V_0 \sin \omega t,$

Given

$$i = 10 \sin 300t \text{ and } v = 200 \sin 300t \text{ V}$$

$$\therefore i_0 = 10 \text{ A and } V_0 = 200 \text{ V}$$

$$\therefore \text{Average power dissipation} = V_0 i_0$$

$$= 200 \times 10 = 2000 \text{ W}$$



**Q4.** Why should the spring / suspension wire in a moving coil galvanometer have low torsional constant?

In moving coil galvanometer, we want to measure even small current also hence for small current galvanometer coil should give deflection.

Current ( $i$ ) and deflection ( $\theta$ ) in Moving coil galvanometer is related by

$$i = \frac{c}{nAB} \theta$$

$n$  = Number of turns in the coil,  $A$  = Area of the coil,  $B$  = magnetic field,  $c$  = torsional couple per unit twist,

Hence  $c$  is kept Low.

Thus to increase sensitivity moving coil galvanometer has low torsional constant (restoring torque per unit twist).

**Q5.** Why does the bluish colour predominate in a clear sky?

We know that scattering is wavelength dependent ( $I \propto \frac{1}{\lambda^4}$ ). Since blue colour has shorter wavelength hence scatters more and sky appears blue.

**Q6.** Which orientation of an electric dipole in a uniform electric field would correspond to stable equilibrium?

Torque on an electric dipole in uniform electric field is  $\tau = \vec{p} \times \vec{E}$

$\vec{p}$  = Electric Dipole Moment

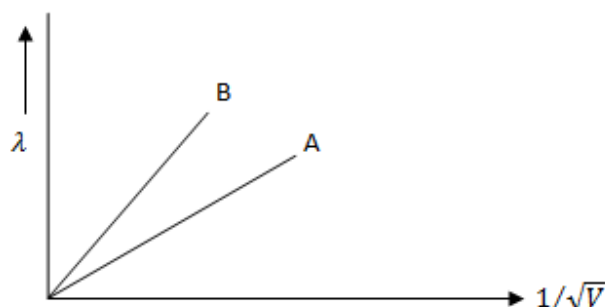
$\vec{E}$  = Electric Field

Cross product is Zero if dipole moment and electric field is in the same direction i.e. parallel.

The dipole is in stable equilibrium when electric dipole is in the direction of electric field.



**Q7.** Two lines, A and B, in the plot given below show the variation of de-Broglie wavelength,  $\lambda$  versus  $1/\sqrt{V}$ , where  $V$  is the accelerating potential difference, for two particles carrying the same charge. Which one of two represents a particle of smaller mass?



We know that wavelength  $\lambda = \frac{h}{\sqrt{2meV}} \Rightarrow \lambda = \frac{h}{\sqrt{m} \cdot \sqrt{2e}} \cdot \frac{1}{\sqrt{V}}$

$$\frac{1}{\sqrt{m}} = \frac{\sqrt{2e}}{h} \cdot \left( \frac{\lambda}{1/\sqrt{V}} \right)$$

$$\therefore \frac{1}{\sqrt{m}} = \frac{\sqrt{2e}}{h} \times (\text{slope of } \lambda \text{ and } 1/\sqrt{V} \text{ graph})$$

$\therefore \text{Slope of B} > \text{Slope of A}$

$$\frac{1}{\sqrt{m_B}} > \frac{1}{\sqrt{m_A}} \Rightarrow \sqrt{m_B} < \sqrt{m_A}$$

$$\therefore m_B < m_A$$

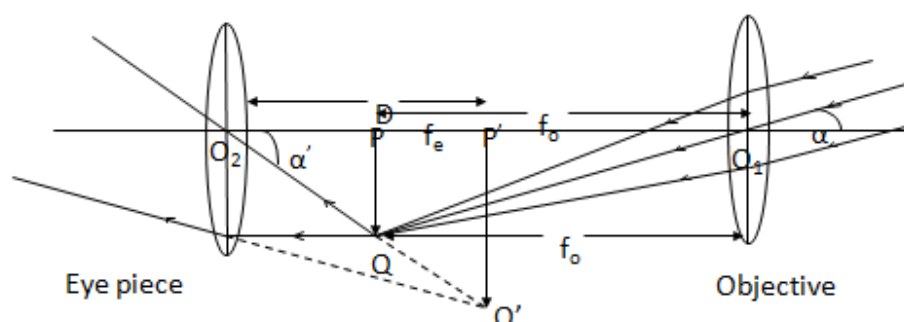
Therefore, line B represents a particle of smaller mass.

**Q8.** State the reason, why Ga As is most commonly used in making of a solar cell ?

The absorption coefficient,  $\alpha$ , is a property of a material which defines the amount of light absorbed by it. Ga As (Gallium Arsenide) has higher absorption coefficient hence most commonly used in making of a solar cell because it absorbs relatively more energy from the incident solar radiations (photon).



**Q9.** Draw a labeled ray diagram of an astronomical telescope in the near point position. Write the expression for its magnifying power.



To find magnifying power ( $M_d$ ):  $O_1P = f_o$ ,  $O_2P = u < f_e$ ,  $O_2P' = D$

$$\therefore \frac{1}{f_e} = \frac{1}{u} + \frac{1}{-D}$$

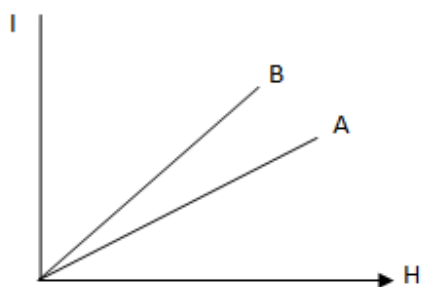
$$\text{or } \frac{1}{u} = \frac{D + f_e}{Df_e}$$

$$\text{For distinct vision } M_d = \frac{\alpha'}{\alpha} = \frac{\tan \alpha'}{\tan \alpha} = \frac{PQ/O_2P}{PQ/O_1P} = \frac{O_1P}{O_2P} = \frac{f_o}{u}$$

$$M_d = f_o \left[ \frac{1}{f_e} + \frac{1}{D} \right] = \frac{f_o}{f_e} \left[ 1 + \frac{f_e}{D} \right]$$



**Q10.** The following figure shows the variation of intensity of magnetization versus the applied magnetic field intensity,  $H$ , for two magnetic materials A and B:



- Identify the materials A and B.
- Why does the material B, have a larger susceptibility than A, for a given field at constant temperature?

Answer:

- Material A is diamagnetic whereas material B is paramagnetic.
- Since paramagnetic substances have a tendency to pull in magnetic field lines when placed in a magnetic field.

**Q11.** Two metallic wires of the same material have the same length but cross-sectional area is in the ratio 1:2. They are connected (I) in series and (II) in parallel. Compare the drift velocities of electrons in the two wires in both the cases (I) and (II).

Given

$$A_1 : A_2 = 1 : 2 \Rightarrow \frac{A_1}{A_2} = \frac{1}{2}$$

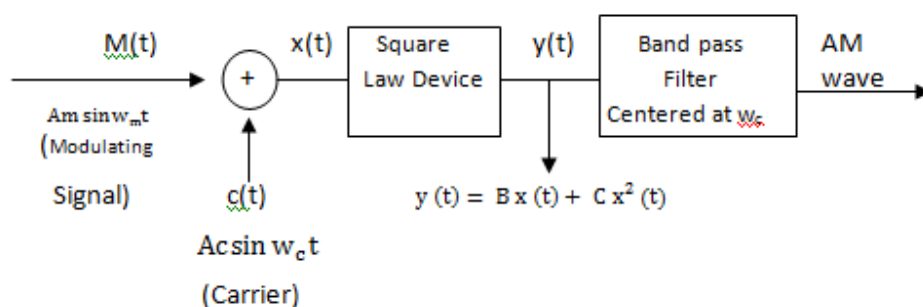
- When two wires are connected in series, the current in both wires A and B will be the same.  
 $\therefore I_A = I_B$   
 $ne A_1 v_{d1} = ne A_2 v_{d2}$   
 $\therefore \frac{v_{d1}}{v_{d2}} = \frac{A_2}{A_1}$   
 $\therefore \frac{v_{d1}}{v_{d2}} = \frac{2}{1} \Rightarrow v_{d1} : v_{d2} = 2 : 1$
- When two wires are connected in parallel, then the potential difference across the wire A and B will be same  
 $V_A = V_B \quad [\because V = IR]$



$$\begin{aligned} \Rightarrow V &= ne A v_d \cdot \rho \frac{l}{A} \\ \therefore V &= ne \rho l v_d \\ \therefore ne \rho l v_{d1} &= ne \rho l v_{d2} \\ \text{or, } v_{d1} &= v_{d2} \\ \therefore \frac{v_{d1}}{v_{d2}} &= \frac{1}{1} \Rightarrow v_{d1} : v_{d2} = 1 : 1 \end{aligned}$$

**Q12.** Draw a block diagram of a simple amplitude modulation. Explain briefly how amplitude modulation is achieved?

Ans. Block diagram to produce AM wave is



In this method the modulating signal  $A_m \sin \omega_m t$  is added to the carrier signal  $A_c \sin \omega_c t$  to produce the signal  $x(t)$ . This signal  $x(t) = A_m \sin \omega_m t + A_c \sin \omega_c t$  is passed through a *square law device* which produces an output.

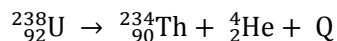
$$y(t) = Bx(t) + Cx^2(t)$$

Where, B and C are constants.

This signal is passed through a *band pass filter* which rejects dc. The output of the band pass filter is therefore, an AM wave.

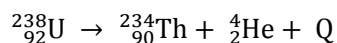


**Q13.** Calculate the energy released in MeV in the following nuclear reaction:



$$\left[ \begin{array}{l} \text{Mass of } {}_{92}^{238}\text{U} = 238.05079 \text{ u} \\ \text{Mass of } {}_{90}^{234}\text{Th} = 234.043630 \text{ u} \\ \text{Mass of } {}_2^4\text{He} = 4.002600 \text{ u} \\ 1 \text{ u} = 931.5 \text{ MeV}/c^2 \end{array} \right]$$

Answer : As per the given nuclear reaction we have



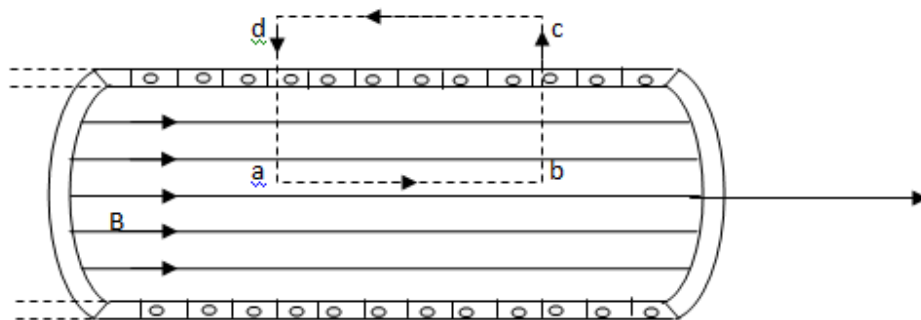
We know that mass defect is the difference in mass between the original atom(Uranium) and constituent atoms ( Thorium + Helium)

$$\begin{aligned} \text{Mass defect} &= M_U - M_{Th} - M_{He} \\ &= 238.05079 - 234.043630 - 4.002600 \\ &= 0.00456 \text{ u} \end{aligned}$$

$$\begin{aligned} \text{Therefore Energy Released} &= 0.00456 \text{ u} \times 931.5 \text{ MeV}/c^2 \\ &= 4.25 \text{ MeV} \end{aligned}$$

**Q14.** Using Ampere's circuital law, obtain an expression for the magnetic field along the axis of a current carrying solenoid of length  $l$  and having  $N$  number of turns.

Ans. **Magnetic field due to solenoid:** Considering a rectangular amperian loop  $abcd$  near the middle of solenoid as shown in the figure. Here length of the Solenoid  $= l$ .



Let the magnetic field along the path  $ab$  be  $B$  and is zero along  $cd$ . As the paths  $bc$  and  $da$  are perpendicular to the axis of solenoid, the magnetic field component along these paths is zero. Therefore, the path  $bc$  and  $da$  will not contribute to the line integral of magnetic field  $B$ .

Total number of turns in length  $l = N/l$ .



The line integral of magnetic field induction  $B$  over the closed path  $abcd$  is

$$\oint_{abcd} \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l} + \int_b^c \vec{B} \cdot d\vec{l} + \int_c^d \vec{B} \cdot d\vec{l} + \int_d^a \vec{B} \cdot d\vec{l}$$

$$\therefore \int_a^b \vec{B} \cdot d\vec{l} = \int_a^b B \, dl \cos 0^\circ = Bl$$

$$\text{and } \int_b^c \vec{B} \cdot d\vec{l} = \int_b^c B \, dl \cos 90^\circ = 0 = \int_d^a \vec{B} \cdot d\vec{l}$$

$$\text{Also } \int_c^d \vec{B} \cdot d\vec{l} = 0 \quad [\because \text{Outside the solenoid, } B = 0]$$

$$\therefore \oint_{abcd} \vec{B} \cdot d\vec{l} = Bl + 0 + 0 + 0 = Bl \dots \dots (i)$$

Using Ampere's circuital law,

$$\oint_{abcd} \vec{B} \cdot d\vec{l} = \mu_0 \times \text{total current in rectangle } abcd$$

$$= \mu_0 \times \text{no. of turns in rectangle} \times \text{current}$$

$$= \mu_0 \times Nl \times I = \mu_0 Nl I \dots \dots (ii)$$

From (i) and (ii), we have

$$Bl = \mu_0 Nl I$$

$$\therefore B = \mu_0 NI$$

**Q15.** Derive an expression for the resistivity of a good conductor in terms of the relaxation time of electrons.

**Answer. Relation between the resistivity & relaxation time:** We know that drift velocity of electron is given by

$$v_d = \frac{eE}{m} \tau$$

$$\text{but } E = \frac{V}{l}$$

$$\therefore v_d = \frac{e}{m} \frac{V}{l} \tau \Rightarrow V = \frac{v_d \cdot ml}{e\tau}$$

We know Ohm's law,

$$R = \frac{V}{I} = \frac{v_d \cdot ml / e\tau}{neA v_d}$$

$$R = \frac{v_d \cdot ml}{e\tau \cdot neA v_d} = \frac{m}{n e^2 \tau} \cdot \frac{l}{A} \dots \dots (1)$$

We know resistivity is given by

$$R = \rho \frac{l}{A} \dots \dots (2)$$





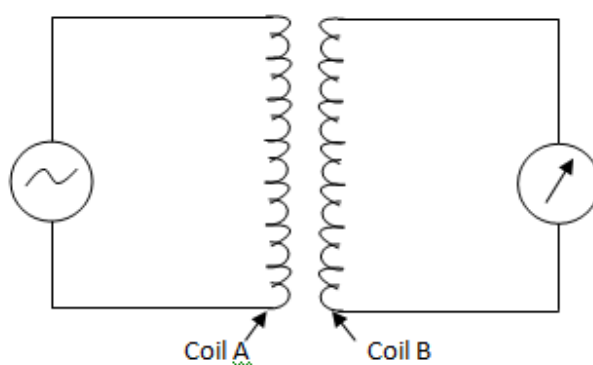
Comparing (1) & (2), we get

$$\rho = \frac{m}{n e^2 \tau}$$

Which is required relationship between resistivity and relaxation time of electrons.

**Q16.** The circuit arrangement given below shows that when an a.c. passes through the coil A, the current starts flowing in the coil B.

- (i) State the underlying principle involved.
- (ii) Mention two factors on which the current produced in the coil B depends.



Answer

- (i) The principle involved in the given circuit arrangement is **mutual induction**. When current flows through the coil, magnetic flux produced cuts the plane of the second coil and it induces emf in the second coil. Mutual induction is the property of two coils by virtue of which each opposes any change in the strength of current flowing through the other by developing an induced emf.
- (ii) The current produced in the coil B depends on
  - a. Number of turns in the coil.
  - b. Nature of material.
  - c. Distance between two coils.



**Q17.** State one feature by which the phenomenon of interference can be distinguished from that of diffraction.

A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 'a'. If the distance between the slits and the screen is 0.8 m and the distance of 2<sup>nd</sup> order maximum from the center of the screen is 15 mm, calculate the width of the slit.

Answer

Diffraction is the bending of waves around an obstacle, while interference is the meeting of two waves. Interference is due to superposition of two distinct waves coming from two coherent sources and Diffraction is produced as a result of superposition of the secondary wavelets coming from different parts of the same wavefront.

**Given**

$$\lambda = 600 \text{ nm} = 600 \times 10^{-9} \text{ m} = 6 \times 10^{-7} \text{ m}$$

$$D = 0.8 \text{ m}, x = 15 \text{ mm} = 1.5 \times 10^{-3} \text{ m}, n = 2, a = ?$$

$$a \frac{x}{D} = n \lambda$$

$$a = \frac{n \lambda D}{x} = \frac{2 \times 6 \times 10^{-7} \times 0.8}{1.5 \times 10^{-3}} = \frac{9.6 \times 10^{-4}}{1.5}$$

$$= 6.4 \times 10^{-4} \text{ m}$$

**Q18.** Two point charges  $q_1 = 10 \times 10^{-8} \text{ C}$  and  $q_2 = -2 \times 10^{-8} \text{ C}$  are separated by a distance of 60 cm in air.

- (I) Find at what distance from the first charge,  $q_1$ , would the electric potential be zero.
- (II) Also calculate the electrostatic potential energy of the system.

Answer



Neutral point must be between the two charges, since Positive charge will repel and negative charge will attract, therefore the direction of force exerted by two charges are equal and opposite.

$$\text{i. Here } q_1 = 10 \times 10^{-8} \text{ C}, q_2 = -2 \times 10^{-8} \text{ C}$$

$$\text{and } AB = 60 \text{ cm} = 0.6 \text{ m}$$

$$\text{Let } AP = x \text{ then } PB = 0.6 - x$$

$$\text{Potential P due to charge } q_1 = \frac{Kq_1}{AP}$$



Potential at P due to charge  $q_2 = \frac{Kq_2}{PB}$

$\therefore$  Potential at P = 0  $\Rightarrow \frac{Kq_1}{AP} + \frac{Kq_2}{PB} = 0$

or,  $\frac{Kq_1}{AP} = -\frac{Kq_2}{PB} \Rightarrow \frac{q_1}{AP} = -\frac{q_2}{PB}$

$\therefore \frac{10 \times 10^{-8}}{x} = \frac{-(-2 \times 10^{-8})}{0.6-x} \Rightarrow \frac{10}{x} = \frac{2}{0.6-x}$

$2x = 6.0 - 10x \Rightarrow 2x + 10x = 6$

$\therefore 12x = 6 \Rightarrow x = \frac{6}{12} = 0.5 \text{ m}$

$\therefore$  Distance from first charge = 0.5 = 50 cm.

(ii) Electrostatic potential energy of the system

$U = K \frac{q_1 q_2}{r}$

$U = 9 \times 10^9 \times \frac{10 \times 10^{-8} \times (-2 \times 10^{-8})}{0.6}$

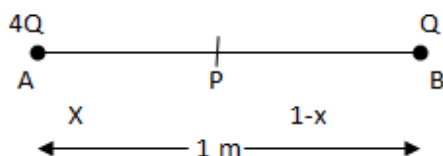
$U = \frac{-18 \times 10^{-6}}{0.6} \Rightarrow U = -30 \times 10^{-6} = -3 \times 10^{-5} \text{ J}$

Or,

Two point charges  $4Q, Q$  are separated by 1m in air. At what point on the line joining the charges is the electric field intensity zero?

Also calculate the electrostatic potential energy of the system of charges, taking the value of charge,  $Q = 2 \times 10^{-7} \text{ C}$  ?

Answer



Let the point be at a distance  $x$  from  $4Q$  charge. Electric field at P due to  $4Q$  = Electric field at P due to  $Q$

$\therefore K \times \frac{4Q}{x^2} = K \times \frac{Q}{(1-x)^2}$



$$\frac{4}{x^2} = \frac{1}{(1-x)^2} \Rightarrow \frac{2}{x} = \pm \frac{1}{1-x}$$

$$\frac{2}{x} = \frac{1}{1-x} \text{ or, } \frac{2}{x} = \frac{-1}{1-x}$$

$$x = 2 - 2x \text{ or, } -x = 2 - 2x$$

$$x + 2x = 2 \text{ or, } -x + 2x = 2$$

$$3x = 2 \text{ or, } x = \frac{2}{3}$$

$$x = \frac{2}{3} \text{ or, } x = 2$$

$\therefore x = 2\text{m}$  is not possible  $\therefore x = \frac{2}{3} \text{ m}$ .

Electrostatic potential energy of the system is

$$U = K \frac{q_1 q_2}{r}$$

$$\Rightarrow U = K \frac{4Q \cdot Q}{r} = K \frac{4Q^2}{r}$$

$$U = 9 \times 10^9 \times \frac{4 \times (2 \times 10^{-7})^2}{1}$$

$$[\because Q = 2 \times 10^{-7} \text{ C}]$$

$$= 9 \times 10^9 \times \frac{4 \times 4 \times 10^{-14}}{1}$$

$$= 144 \times 10^{-5} = 1.44 \times 10^{-3} \text{ J}$$

**Q19.** Identify the following electromagnetic radiations as per the wavelengths given below. Write one application of each.

(a)  $10^{-3} \text{ nm}$

(b)  $10^{-3} \text{ m}$

(c)  $1 \text{ nm}$

Ans.

(a) **Wavelength range :**  $10^{-3} \text{ nm} = 10^{-3} \times 10^{-9} = 10^{-12} \text{ m}$

**Type of EMW:** X-rays

**Application :**

X-rays reveal structural information about the material through which it passes or falls over. X-rays are widely used in medicine to reveal the architecture of the bone and other soft tissues and to find out any abnormality in the form of fracture, growth of tumor etc. It is also used in dental imaging. X-rays in airports to examine for the presence of dangerous weapons or bombs



(b) **Wavelength range:**  $10^{-3}$  m

**Type of EMW:** Microwave

**Application:** Microwaves are used in Radar systems for air craft navigation and for cooking.

(c) **Wavelength range:** 1 nm

**Type of EMW:** Infra-red

**Application:** Infra-red waves are used for taking photographs during the conditions of fog, smoke etc. Doctors use infrared lamps to treat skin diseases and relieve the pain

**Q20.** Explain why high frequency carrier waves are needed for effective transmission of signals.

A message signal of 12 kHz and peak voltage 30 V. Calculate the (i) modulation index, (ii) side-band frequencies.

Answer

For the effective transmission of signals, the high frequency carrier waves are used because these high frequency carrier waves travel through space or medium with the speed of light and they are not obstructed by earth's atmosphere.

**Numerical:**  $v_m = 12$  kHz,  $E_m = 20$  V,  $v_c = 12$  MHz = 12000 kHz,  $E_c = 30$  V

$$\text{Modulation index, } \mu_a = \frac{A_m}{A_c} = \frac{20}{30} = 0.67$$

(ii) The side bands are

$$\text{USB} = v_c + v_m = 12000 + 12 = 12012 \text{ kHz}$$

$$\text{LSB} = v_c - v_m = 12000 - 12 = 11988 \text{ kHz}$$



**Q21.** Distinguish between unpolarized and plane polarized light. An unpolarized light is incident on the boundary between two transparent media. State the condition when the reflected wave is totally plane polarized. Find out the expression for the angle of incidence in this case.

Answer

For unpolarized light, light waves vibrate in more than one plane. Therefore independent light waves whose planes of vibrations are randomly oriented about the direction of propagation are said to be **unpolarized**.

For Polarized light, vibrations of light waves occur in a single plane. When the light waves transmit only one component parallel to a special axis, then the resulting light is called **plane polarized** or **linearly polarized** light.

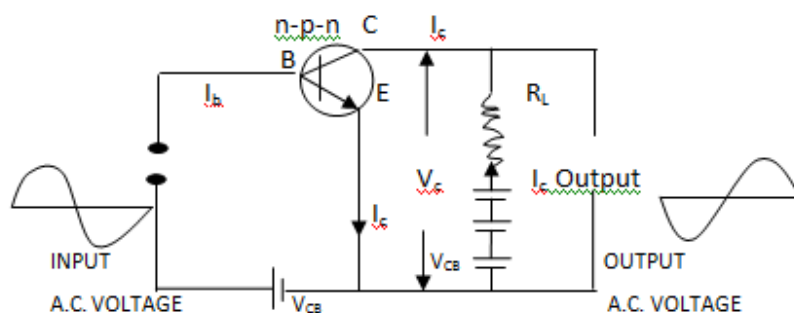
When reflected wave is perpendicular to the refracted wave, the reflected wave is totally polarized wave. The angle of incidence in this case is called **Brewster's angle** and is denoted by  $i_B$ .

The relation with incident angle  $i_B$  is given by  $\mu = \tan i_B$

**Q22.** Draw the labeled circuit diagram of a common-emitter transistor amplifier. Explain clearly how the input and output signals are in opposite phase.

Answer

The circuit details for using an npn transistor as common emitter amplifier are shown in the figure.



The input (base-emitter) circuit is forward biased and the output (collector-emitter) circuit is reversed biased. When no a.c. signal is applied, the potential difference  $V_C$  between the collector and emitter, is given by

$$V_C = V_{ce} - I_C \times R_L \dots \dots (1)$$

Where  $V_{ce}$  is the voltage of battery  $V_{CE}$



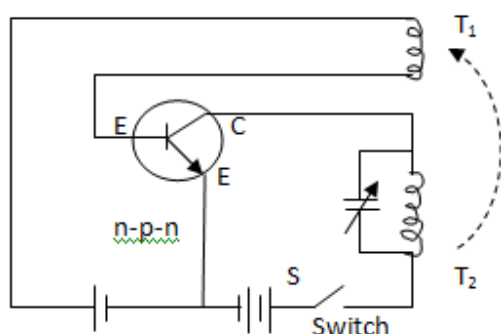
When an a.c. signal is fed to the input circuit, the forward bias increases during the positive half cycle of the input. This results in an increase in  $I_C$  and a consequent decrease in  $V_C$ , as is clear from (1). Thus during positive half cycle of the input, the collector becomes less positive.

During the negative half cycle of the input, the forward bias is decreased resulting in a decrease in  $I_E$  and hence  $I_C$ . Therefore, from (1)  $V_C$  would increase, making the collector more positive. *Hence in a common-emitter amplifier, the output voltage is  $180^\circ$  out of phase with the input voltage.*

**Q22(II)** State briefly the underlying principle of a transistor-oscillator. Draw a circuit diagram showing how the feedback is accomplished by inductive coupling. Explain the oscillator action.

Answer

**Transistor as an oscillator:** In an oscillator, the output at a desired frequency is obtained without applying any external input voltage. The common emitter npn transistor as an oscillator is shown in the figure. A variable capacitor  $C$  of suitable range is connected in parallel to coil  $L$  to give the variation in frequency.



**Oscillator action:** As in an amplifier, the base-emitter junction is forward biased while the base collector junction is reversed biased. When the switch  $S$  is put on, a surge of collector current flows in the coil  $T_2$ . The inductive coupling between coil  $T_2$  and  $T_1$  cause a current to flow in the emitter circuit i.e., feedback from input to output. As a result of positive feedback, the collector current reaches at maximum. When there will be no further feedback from  $T_2$  to  $T_1$ , the emitter current begins to fall and collector current decreases. Therefore, the transistor has reverted back to its original state. The whole process now repeats itself.

The resonance frequency ( $f$ ) of the oscillator is given by

$$f = \frac{1}{2\pi\sqrt{LC}}$$

The tank of tuned circuit is connected in the oscillator side. Hence it is known as *tuned collector oscillator*.



**Q23.** The ground state energy of hydrogen atom is -13.6 eV.

- (I) What is the kinetic energy of an electron in the 2<sup>nd</sup> excited state?
- (II) If the electron jumps to the ground state from the 2<sup>nd</sup> excited state, calculate the wavelength of the spectral line emitted.

**Answer**

The energy of an electron in nth orbit is given by

$$E_n = \frac{-13.6}{n^2} \text{ eV}$$

- (I) For 2<sup>nd</sup> excited state,  $n = 3$

$$\therefore E_3 = \frac{-13.6}{3^2} = \frac{-13.6}{9} = -1.51 \text{ eV}$$

- (II) Required energy to jump electron to the ground state from the 2<sup>nd</sup> excited state  
 $E = E_3 - E_1$

$$= \frac{-13.6}{(3)^2} - \left( \frac{-13.6}{(1)^2} \right) = -1.51 + 13.6 = 12.09 \text{ eV}$$

$\therefore$  Wavelength of the spectral line emitted is

$$\lambda = \frac{hc}{E} \quad \left[ \because E = \frac{hc}{\lambda} \right]$$

$$\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{12.09 \times 1.6 \times 10^{-19}}$$

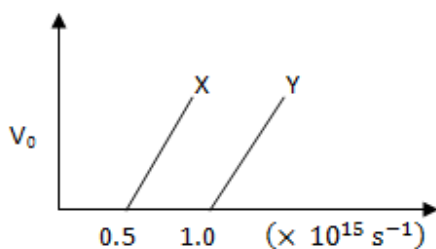
$$= \frac{19.878 \times 10^{-7}}{19.344}$$

$$= 1.027 \times 10^{-7} \text{ m} = 1027 \text{ \AA}$$





**Q24.** The following graph shows the variation of stopping potential  $V_0$  with the frequency  $\nu$  of the incident radiation for two photosensitive metals X and Y:



Which of the metal has larger threshold wavelength? Give reason.

- (i) Explain, giving reason, which metal gives out electrons, having larger kinetic energy, for the same wavelength of the incident radiation.
- (ii) If the distance between the light source and metal X is halved, how will the kinetic energy of electrons emitted from it change? Give reason.

Ans.

(i) Let  $\nu_0$  and  $\nu'_0$  be the Frequencies of incident radiations of metal Y and X respectively.

$$\therefore \nu_0 > \nu'_0$$

$$\therefore \nu_0 = \frac{c}{\lambda_0} \therefore \frac{c}{\lambda_0} > \frac{c}{\lambda'_0} \Rightarrow \lambda_0 < \lambda'_0$$

Therefore, metal X has larger threshold wavelength.

(ii) Since  $E = h \nu_0 \Rightarrow E \propto \nu_0$

Therefore, metal Y has larger kinetic energy.

(iii)  $E = e V_0$

Therefore, kinetic energy of the emitted photoelectron is independent of the intensity of the incident light, hence K.E. of the emitted photoelectrons remains unchanged if the intensity of the incident radiation is halved.



**Q25.** A circular coil of 200 turns and radius 10 cm is placed in a uniform magnetic field of 0.5 T, normal to the plane of the coil. If the current in the coil is 3.0 A, calculate the

- (i) Total torque on the coil.
- (ii) Total force on the coil.
- (iii) Average force on each electron in the coil, due to the magnetic field.

Assume the area of cross-section of the wire to be  $10^{-5} \text{ m}^2$  and the free electron density is  $10^{29}/\text{m}^3$ .

Answer

Given  $n = 200$ ,  $r = 10 \text{ cm} = 0.1 \text{ m}$ ,  $B = 0.5 \text{ T}$ ,  $\alpha = 0^\circ$  and  $I = 3.0 \text{ A}$

Area of the coil  $A = \pi r^2 = 3.14 \times (0.1)^2$

$$\therefore A = 3.14 \times 0.01 = 0.0314 \text{ m}^2$$

$$(i) \tau = nIBA \sin \alpha = nIBA \sin 0^\circ = 0$$

(ii) Total force on a planar current loop in a magnetic field is always zero.

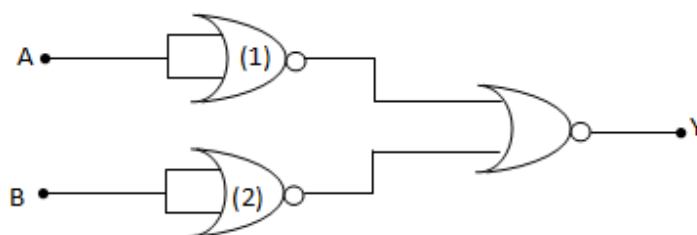
$$(iii) \text{ Given } N = 10^{29}/\text{m}^3, A = 10^{-5} \text{ m}^2$$

Average force on an electron of charge ( $e$ ), moving with drift velocity ( $v_d$ ) in the magnetic field ( $B$ ) is given by

$$F = Be v_d = Be \frac{I}{neA} \quad [\because I = ne Av_d]$$

$$F = \frac{BI}{nA} = \frac{0.5 \times 3.0}{10^{29} \times 10^{-5}} = 1.5 \times 10^{-24} \text{ N.}$$

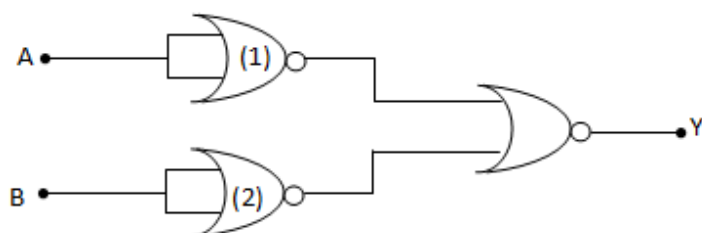
**Q26.** The inputs A and B are inverted by using two NOT gates and their outputs are fed to the NOR gate as shown below.



Analyze the action of the gates (1) and (2) and identify the logic gate of the complete circuit so obtained. Give its symbol and the truth table.

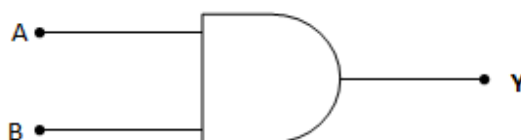


Answer



The output  $Y = \overline{\overline{A}} + \overline{\overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B$

- (i) The gates (1) and (2) act as NOT gate.
- (ii) The logic gate of the complete circuit is AND gate.

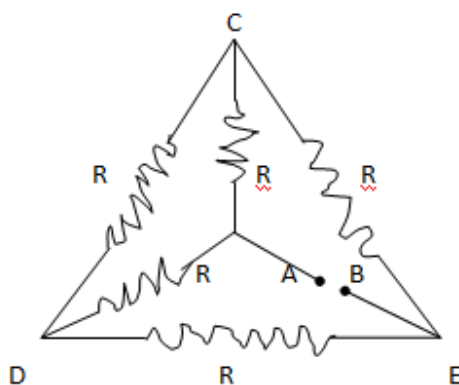


Truth Table

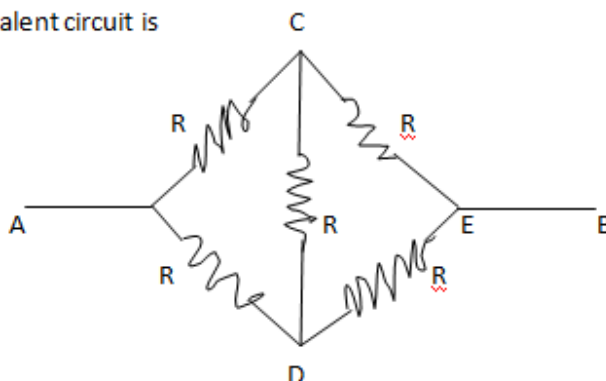
| Input |   | Output<br>$Y = A + B$ |
|-------|---|-----------------------|
| A     | B |                       |
| 0     | 0 | 0                     |
| 0     | 1 | 0                     |
| 1     | 0 | 0                     |
| 1     | 1 | 1                     |



- Q27.** (i) Calculate the equivalent resistance of the given electrical network between points A and B.  
 (ii) Also calculate the current through CD and ACB, if a 10 V d.c. source is connected between A and B, and the value of R is assumed as  $2\ \Omega$ .



Ans. (i) Equivalent circuit is



Which is a balanced *Wheatstone bridge*. Therefore, the resistance of arm CD is ineffective.

Thus resistance of arm ACB,  $R_1 = R + R = 2R$  and resistance of arm ADB,  $R_2 = R + R = 2R$

$\therefore$  Effective resistance between A and B is

$$\frac{1}{R_p} = \frac{1}{2R} + \frac{1}{2R} \Rightarrow \frac{1}{R_p} = \frac{2}{2R}$$

$$\therefore R_p = R$$

(ii)  $\therefore$  resistance of arm CD is ineffective

$\therefore$  No current flows in arm CD

Resistance of arm ACB =  $2R = 2 \times 2 = 4\ \Omega$

$$\therefore \text{Current through ACB, } I = \frac{V}{R} = \frac{10}{4} = 2.5\ \text{A}$$



**Q28.** Derive an expression for the energy stored in a parallel plate capacitor.

On charging a parallel plate capacitor to a potential  $V$ , the spacing between the plates is halved, and a dielectric medium of  $\epsilon_r = 10$  is introduced between the plates, without disconnecting the d.c. source. Explain, using suitable expressions, how the (i) capacitance, (ii) electric field and (iii) energy density of the capacitor change.

Ans. **Energy stored in a parallel plate capacitor:** Work is done in charging a capacitor. This work done is stored as its electrical potential energy. Suppose a capacitor is charged with charge  $q$  show that potential difference between its plates is

$$V = \frac{q}{C}$$

Work done to increase the charge by an amount  $dq$  is

$$dW = V dq = \frac{q}{C} dq$$

Total work done to charge the capacitor from 0 to  $Q$  is

$$W = \int_0^Q \frac{q}{C} dq = \frac{1}{C} \left[ \frac{q^2}{2} \right]_0^Q = \frac{Q^2}{2C}$$

$$\therefore \text{Energy of the capacitor, } U = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV \quad \left[ \because C = \frac{Q}{V} \right]$$

$$\text{i. Here } K = \epsilon_r = 10 \quad \therefore C = \frac{\epsilon_0 A}{d}$$

$$\therefore C_1 = K \frac{\epsilon_0 A}{d/2} = 10 \times 2 \times \frac{\epsilon_0 A}{d} = 20 C$$

$\therefore$  Capacitance becomes 20 times.

(ii) Electric field between the plates remains unchanged ( $E_1 = E$ ) because the p.d. across the plates remains unchanged.

$$\text{iii. } U = \frac{1}{2} CV^2$$

$$\therefore U_1 = \frac{1}{2} C_1 V^2 = \frac{1}{2} \times 20C \times V^2$$

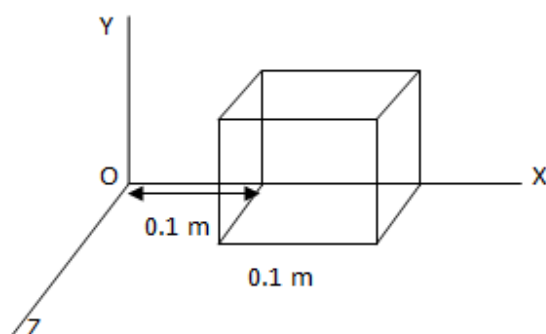
$$= 20 \left( \frac{1}{2} C V^2 \right) = 20 U$$

$\therefore$  Energy stored increases 20 times.



**Q28 – Part II**

- (a) Define electric flux. Write its SI unit.  
 (b) The electric field components due to a charge inside the cube of side 0.1 m are as shown:



$$E_x = \alpha x, \text{ where } \alpha = 500 \text{ N/C} - \text{m}$$

$$E_y = 0, E_z = 0.$$

Calculate (i) the flux through the cube, and (ii) the charge inside the cube.

Answer

(a) **Electric flux:** Electric flux over an area in an electric field represents the total number of electric field lines crossing this area. It is denoted by  $\phi_E$ .

Electric flux is a scalar quantity; its SI unit is  $\text{Nm}^2 \text{C}^{-1}$ .

(b)(i) This cube consists of 6 faces, we know that flux is dot product of Electric Field and Area.

Along X axis lines of force will only cut faces vertical to X axis i.e only left and right faces other faces are parallel to X axis hence flux will be Zero.

$$E_x = \alpha x = \alpha a \quad \text{Therefore Flux } \phi_L = \vec{E} \cdot \vec{\Delta S} = \alpha a \cdot a^2 \cos 180^\circ = -\alpha a^3$$

For the right vertical face of the cube

$$E_x = \alpha x = \alpha 2a \quad \text{Therefore Flux } \phi_R = \vec{E} \cdot \vec{\Delta S} = \alpha 2a \cdot a^2 \cos 0^\circ = 2\alpha a^3$$

$$\text{Therefore total flux} = \phi = \phi_L + \phi_R = -\alpha a^3 + 2\alpha a^3 = \alpha a^3 = 500 \times (0.1)^3 = 0.5 \text{ Nm}^2 \text{C}^{-1}$$

(ii) By Gauss's law

$$q = \epsilon_0 \phi$$

$$= 8.85 \times 10^{-12} \times 0.5 = 4.425 \times 10^{-12} \text{C}$$

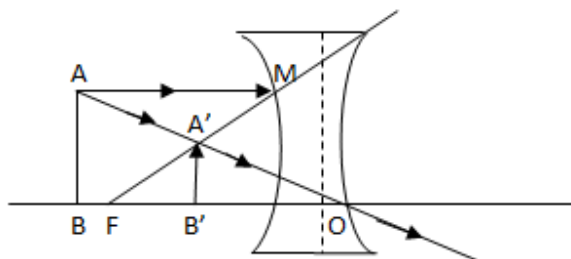


**Q29.** Derive the lens formula,  $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$  for a concave lens, using the necessary ray diagram.

Two lenses of powers 10 D and -5 D are placed in contact.

- (i) Calculate the power of the new lens.
- (ii) Where an object should be held from the lens, so as to obtain a virtual image of magnification 2?

**Derivation of lens formula:** In the figure, AB is the object kept beyond F perpendicular to the principal of a concave lens. A'B' is the erect, virtual and diminished image.



$\Delta ABO$  and  $\Delta A'B'O$  are similar.

$$\therefore \frac{AB}{A'B'} = \frac{BO}{B'O} \dots \dots \dots (i)$$

$\Delta MOF$  and  $\Delta A'B'F$  are similar.

$$\frac{MO}{A'B'} = \frac{FO}{FB'}$$

But  $MO = AB$

$$\frac{AB}{A'B'} = \frac{FO}{FB'} \dots \dots \dots (ii)$$

From equations (i) and (ii), we get

$$\frac{BO}{B'O} = \frac{FO}{FB'} \text{ or, } \frac{BO}{B'O} = \frac{FO}{FO - B'O}$$

Using new Cartesian sign convention,

$$BO = -u, B'O = -v, FO = -f$$

$$\text{Hence } \frac{-u}{-v} = \frac{-f}{-f+v}$$

$$\text{Or, } uf - uv = vf$$

$$\text{Or, } uv = uf - vf$$

Dividing both sides by  $uvf$ , we have



$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

This is the required formula for a thin concave lens.

- (i) Power of new lens,  $P = P_1 + P_2$   
 $\therefore P = 10 - 5 = +5D$
- (ii) Here,  $u = ?$ ,  $f = \frac{1}{P} = \frac{100}{5} = 20 \text{ cm}$   
 $m = \frac{v}{u} \Rightarrow 2 = \frac{-v}{-u} \therefore v = 2u$

Using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{2u} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1-2}{2u} \Rightarrow \frac{1}{20} = \frac{-1}{2u} \therefore u = -10 \text{ cm.}$$

$\therefore$  Object distance = 10 cm.

#### Q29 Part II

- (a) What are the coherent sources of light? Two slits in Young's double slit experiment are illuminated by two different sodium lamps emitting light of the same wavelength. Why is no interference pattern observed?
- (b) Obtain the condition for getting dark and bright fringes in Young's experiment. Hence write the expression for the fringe width.
- (c) If  $s$  is the size of the source and its distance from the plane of the two slits, what should be the criterion for the interference fringes to be seen?

Answer

- (a) **Coherent sources of light:** The sources of light, which emits continuous light waves of the same wavelength, same frequency and in same phase are called **Coherent sources of light**. Coherent sources are always produced from a single source in two different ways.
- ( 1 ) By division of wave front : We know any point laying on the wave front act as secondary coherent sources. We have to expose only two such points blocking all other points as done in Young's experiment.
- ( 2 ) By division of amplitude : We know that intensity of light is proportional to square of the amplitude.





Interference pattern is not obtained. This is because phase difference between the light waves emitted from two different sodium lamps will change continuously.

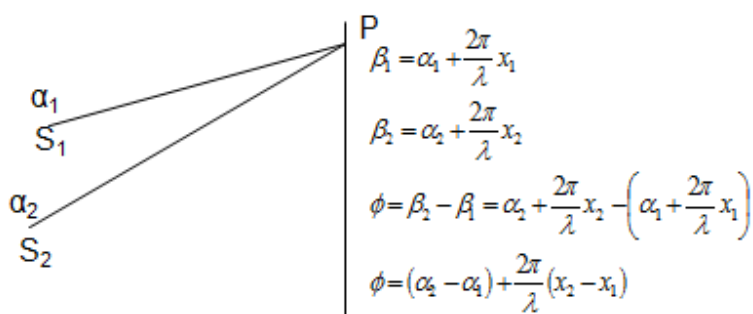
(b)

Let  $d$  = distance of separation between two slits.

$D$  = Perpendicular distance of the screen from the slits

$a$  &  $R$  = amplitude of the waves emitted from  $S_1$  and  $S_2$  respectively.

$\lambda$  = wave length of light ( $\omega = 2\pi\nu = 2\pi c/\lambda$ )



Let  $y_1$  and  $y_2$  be the displacement at any point  $P$  on the screen at an instant of time  $t$  of the waves emitted from  $S_1$  and  $S_2$  respectively. Using displacement equation:

$$y_1 = a \sin \omega t \rightarrow (1)$$

$$y_2 = a \sin (\omega t \pm \phi) \rightarrow (2)$$

$$\phi = (\alpha_2 - \alpha_1) + \frac{2\pi}{\lambda} (x_2 - x_1) \rightarrow (3)$$

$$y = y_1 + y_2 = a \sin \omega t + a \sin (\omega t \pm \phi)$$

$$y = a \sin \omega t + b \sin \omega t \cos \phi \pm b \cos \omega t \sin \phi$$

$$y = (a + b \cos \phi) \sin \omega t \pm b \cos \omega t \sin \phi \rightarrow (4)$$

$$\text{Putting } b \sin \phi = c \sin \delta \rightarrow (5)$$

$$a + b \cos \phi = c \cos \delta \rightarrow (6)$$

Putting equation (5) and (6) in (4)

$$y = c \sin \omega t \cos \delta \pm c \cos \omega t \sin \delta$$

$$y = c \sin (\omega t \pm \delta) \rightarrow (7)$$



From equation (7) we find that  $c$  is the amplitude of the resultant wave due to the superposition. Squaring and adding equation (5) & (6)

$$c^2 = b^2 \sin^2 \phi + a^2 + b^2 \cos^2 \phi + 2ab \cos \phi$$

$$c^2 = a^2 + b^2 + 2ab \cos \phi \rightarrow (8)$$

$I$  = Resultant intensity at P due to the superposition of the two waves

$$I \propto (\text{amplitude})^2$$

$$I \propto c^2$$

$$I = \text{constant} \cdot c^2$$

$$I = \text{constant} \cdot [a^2 + b^2 + 2ab \cos \phi] \rightarrow (9)$$

From equation (9) we find  $I$  is maximum when  $\cos \phi$  is maximum.

$$I = \text{constant} [a^2 + b^2 + 2ab \cos \phi]$$

$$\cos \phi = 1 = \cos 0 = \cos 2\pi = \cos 4\pi = \cos 6\pi = \dots$$

$$\phi = 2n\pi \rightarrow (10) \text{ where } n = 0, 1, 2, 3, 4, \dots$$

$$\text{or } \frac{2\pi}{\lambda} \Delta = 2n\pi \rightarrow (11)$$

$$\Delta = \frac{2n\pi}{2\pi} \lambda = 2n \frac{\lambda}{2} \rightarrow (12)$$

where  $n = 0, 1, 2, 3, \dots$  an integer

$$\Delta = 0, \lambda, 2\lambda, 3\lambda, 4\lambda, \dots$$

Condition of minimum intensity (Destructive interference): From equation (9) intensity is minimum when  $\cos \phi$  is minimum i.e.  $\cos \phi = -1 = \cos \pi = \cos 3\pi = \dots$

i.e.  $\phi = \pi, 3\pi, 5\pi, 7\pi, \dots$

$$\phi = (2n+1)\pi \rightarrow (13)$$

where  $n = 0, 1, 2, 3, 4, \dots$



$$\therefore \frac{2\pi}{\lambda} \Delta = (2n + 1)\pi$$

$$\text{or } \Delta = (2n + 1) \frac{\lambda}{2} \rightarrow (14)$$

Where  $n = 0, 1, 2, 3, \dots$  .. an integer

$$\text{i.e. } \Delta = 1. \frac{\lambda}{2}, 3 \frac{\lambda}{2}, 5 \frac{\lambda}{2}, 7 \frac{\lambda}{2}, \dots$$

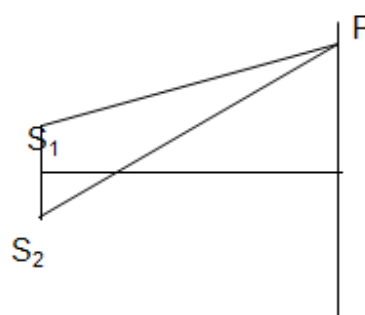
**Fringe width** : The distance of separation between any two consecutive bright or dark fringes is known as fringe width.

The constructive interference happens when wavelets emitting from both sources are in phase

$$\text{i.e. } S_2P - S_1P = n\lambda; n = 0, 1, 2, \dots$$

Then

$$S_2P = \left( D^2 + \left( x + \frac{d}{2} \right)^2 \right)^{\frac{1}{2}} \approx D \left[ 1 + \frac{1}{2} \left( \frac{x + \frac{d}{2}}{D} \right)^2 \right]$$



$$S_1P = D \left[ 1 + \frac{1}{2} \left( \frac{x - \frac{d}{2}}{D} \right)^2 \right]$$

Therefore, path difference

$$S_2P - S_1P = \frac{\left( x + \frac{d}{2} \right)^2 - \left( x - \frac{d}{2} \right)^2}{2D} = \frac{xd}{D}$$

Therefore, the point P will be maximum intensity if

$$\frac{xd}{D} = n\lambda$$

or

$$x = \frac{n\lambda D}{d}$$



The condition for destructive interference is

$$\text{path difference} = \left(n + \frac{1}{2}\right)\lambda$$

Therefore,

$$\frac{xd}{D} = \left(n + \frac{1}{2}\right)\lambda$$

or

$$x = \frac{(2n+1)D\lambda}{2d}$$

The fringe width is the separation between the consecutive bright or dark fringes

Taking bright fringes

$$\beta = x_{n+1} - x_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d} = \frac{\lambda D}{d}$$

(c) The conditions for interference fringes to be seen is

(1)

$$I_{\max} = \text{constant} \left[ a^2 + b^2 + 2ab \right]$$

$$I_{\max} = \text{constant} (a + b)^2 \rightarrow (15)$$

$$I_{\min} = \text{constant} \left[ a^2 + b^2 + 2ab(-1) \right]$$

$$I_{\min} = \text{constant} (a - b)^2 \rightarrow (16)$$

If  $a=b$  i.e. if the amplitude of the two interfering waves are equal then from equation

(15) and (16) :  $I_{\max} = \text{constant} \cdot 4a^2$  ,  $I_{\min} = 0$ . Contrast between the bright and dark fringe is very high and the fringes are very good.



If  $a \gg b$  then  $b$  can be neglected compared to  $a$  and hence from (15) and (16)  $I_{\max} = \text{constant} \cdot a^2$ ,  
 $I_{\min} = \text{constant} \cdot a^2$  so fringes can not be seen.

*Thus to get good interference fringes the amplitude of the two interfering waves must be either exactly equal or very nearly equal. But if they are widely different no fringes can be seen.*

(2)

$$D = 1 \text{ m} = 100 \text{ cm}$$

$$\lambda = 6000 \text{ \AA} = 6 \times 10^{-5} \text{ cm. then Fringe width}$$

$$x = 0.006 \text{ cm}$$

We find that the fringe width is so small that bright and dark fringes will overlap. Hence to have distinct interference fringes the distance of separation between the two coherent sources should be very small, smaller than 1mm.

(3) From Equation 9 we find that resultant intensity

$$I = \text{constant} \cdot [a^2 + b^2 + 2ab \cos \phi] \rightarrow (9)$$

$$\phi = (\alpha_2 - \alpha_1) + \frac{2\pi}{\lambda}(x_2 - x_1)$$

If the sources are not coherent the initial phase difference ( $\alpha_2 - \alpha_1$ ) changes with time  $10^8$  times per second, i.e. intensity at every point changes  $10^8$  times per second.

Since human eye can not follow such quick change hence intensity at every point appears to be same i.e. no fringes can be seen. But if the sources are coherent initial phase are same  $\alpha_2 - \alpha_1 = 0$ . path difference is different at different points on the screen hence the phase difference  $\phi$  and so the intensity  $I$  are different at different points on the screen and we get interference fringes.



**Q30.** An a.c. source generating a voltage  $v = v_m \sin \omega t$  is connected to a capacitor of capacitance  $C$ . Find the expression for the current,  $i$ , flowing through it. Plot a graph of  $v$  and  $i$  versus  $\omega t$  to show that the current is  $\pi/2$  ahead of the voltage.

A resistor of  $200 \, \Omega$  and a capacitor of  $15.0 \, \mu\text{F}$  are connected in series to a  $220 \, \text{V}$ ,  $50 \, \text{Hz}$  a.c. source. Calculate the current in the circuit and the rms voltage across the resistor and the capacitor. Is the algebraic sum of these voltages more than the source voltage? If yes, resolve the paradox.

Ans. **A.C. source containing capacitor:** Let a source of alternating emf  $V = V_m \sin \omega t$  be connected to a capacitor of capacitance  $C$  only.

$$V = V_m \sin \omega t \dots \dots \dots (1)$$

At every instant, the potential  $V$  is given by

$$V = \frac{q}{c} \Rightarrow V_m \sin \omega t$$

$$\therefore q = C V_m \sin \omega t$$

If  $i$  is instantaneous value of current in the circuit at instant  $t$ , then

$$i = \frac{dq}{dt} = \frac{d}{dt} (C V_m \sin \omega t)$$

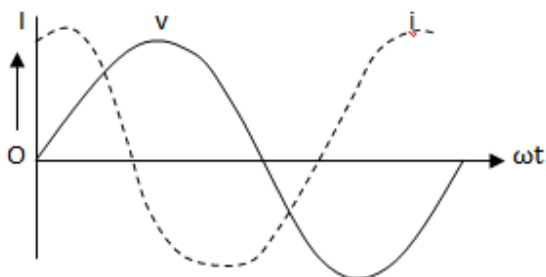
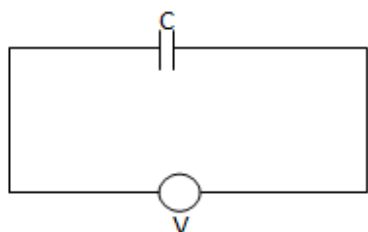
$$i = C V_m (\cos \omega t) \cdot \omega = \frac{V_m}{1/\omega c} \sin \left( \omega t + \frac{\pi}{2} \right)$$

The current will be maximum when  $\sin \left( \omega t + \frac{\pi}{2} \right) = 1$

$$\therefore i_m = \frac{V_m}{1/\omega c} \times 1 = \frac{V_m}{1/\omega c}$$

$$i = i_m \sin \left( \omega t + \frac{\pi}{2} \right)$$

Therefore, alternating current  $i$  lead the alternating voltage by a phase angle of  $\frac{\pi}{2}$ .



Given  $R = 200 \, \Omega$ ,  $C = 15.0 \, \mu\text{F} = 15 \times 10^{-6} \, \text{F}$ ,

$V_{\text{rms}} = 220 \, \text{V}$ ,  $\nu = 50 \, \text{Hz}$ ,  $I_{\text{rms}} = ?$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$$

$$= \frac{1}{2 \times \frac{22}{7} \times 50 \times 15 \times 10^{-6}} = \frac{7 \times 10^6}{33000}$$

$$= 212.12 = 212 \, \Omega$$

$$\therefore Z = \sqrt{R^2 + X_C^2} = \sqrt{200^2 + 212^2}$$

$$= \sqrt{40000 + 44944} = \sqrt{84944} = 291.45 \, \Omega$$

$$\therefore I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{220}{291.45} = 0.75 \, \text{A}$$

$$\therefore V_R = I_{\text{rms}} R = 0.75 \times 200 = 150 \, \text{V}$$

$$V_C = I_{\text{rms}} \cdot X_C = 0.75 \times 212 = 159 \, \text{V}$$

$$\therefore V_R + V_C = 150 + 159 = 309$$

$$\therefore V_R + V_C > V$$

This (paradox) is because these voltages **are not in same phase** and they can not be added like ordinary numbers.



$$\therefore V = \sqrt{V_R^2 + V_C^2}$$

$$= \sqrt{150^2 + 159^2} = \sqrt{22500 + 25281}$$

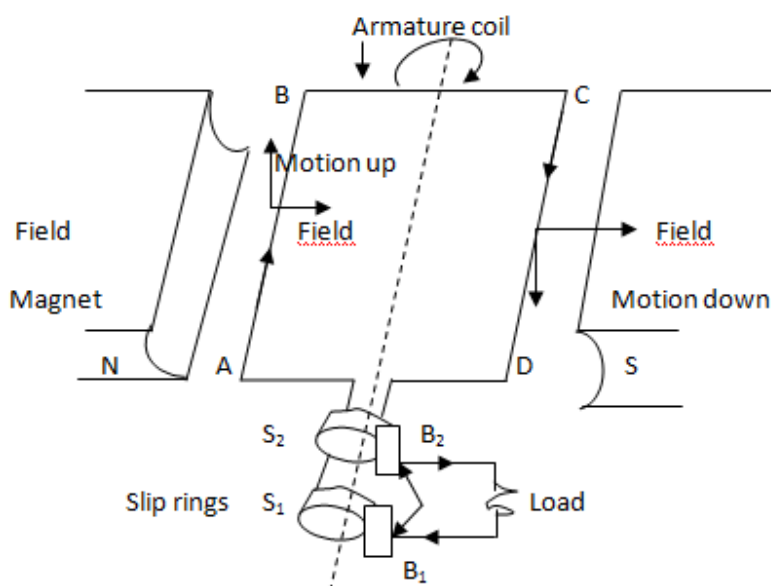
$$= \sqrt{47781} = 218.58 \text{ V}$$

Or,

Explain briefly, with the help of a labeled diagram, the basic principle of the working of an a.c. generator. In an a.c. generator, coil of  $N$  turns and area  $A$  is rotated at  $v$  revolutions per second in a uniform magnetic field  $B$ . write the expression for the emf produced.

A 100-turn coil of area  $0.1 \text{ m}^2$  rotates at half a revolution per second. It is placed in a magnetic field  $0.01 \text{ T}$  perpendicular to the axis of rotation of the coil. Calculate the maximum voltage generated in the coil.

**Principle Of A. C. Generator** : Whenever a closed coil is rotated in a uniform magnetic field about an axis perpendicular to the field, the magnetic flux linked with coil changes and an induced emf is set up across its ends.







The essential parts of an a.c. generator are shown in the figure. Initially the armature coil ABCD is horizontal. As the coil is rotated clockwise, the arm AB moves up and CD moves down. By Fleming's right hand rule, the induced current flows along ABCD. In second half rotation, the arm CD moves up and AB moves down. The induced current flows in the opposite direction i.e., along DCBA. Thus an alternating current flows in the circuit.

The magnetic flux linked with the coil at any instant is

$$\phi = NB A \cos \omega t$$

Induced emf will be

$$E = - \frac{d\phi}{dt} = - \frac{d}{dt} (NB A \cos \omega t)$$

$$= NBA \omega \sin \omega t$$

$$\text{or, } E = E_0 \sin \omega t$$

Where  $E_0 = NBA \omega = \text{peak value of induced emf.}$

**Solution of Numerical Problem:** Given  $N = 100, A = 0.1 \text{ m}^2, B = 0.01 \text{ T}$

$$\nu = \frac{1}{2} \text{ revolution per sec} = 0.5 \text{ r. p. s}$$

$$\therefore \text{Maximum voltage generated } e_0 = NBA \omega = NBA (2\pi \nu)$$

$$\therefore e_0 = 100 \times 0.01 \times 0.1 \times 2 \times \frac{22}{7} \times 0.5$$

$$= \frac{2.2}{7} = 0.314 \text{ Volt}$$

$$e_{\text{rms}} = \frac{e_0}{\sqrt{2}} = \frac{0.314}{1.414} = 0.22 \text{ V.}$$